Zilionis 2019 Non-Small-Cell Lung Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥131 cells) intermix on one continuous manifold with no per-patient / per-histology islands (radius-graph: 0 detached; iLISI median 1.00→3.35 of 6); both Leiden lobes carry all 6 samples and both histologies (LUAD+LUSC). Read on the embedding, not a scalar alone.

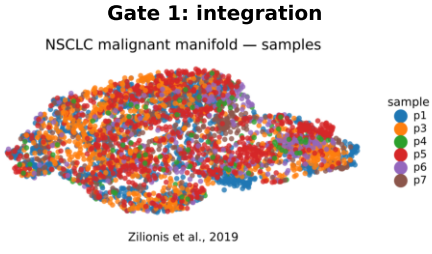
Gate 2 — module geography → GO (non-branching, Crest-flagged)

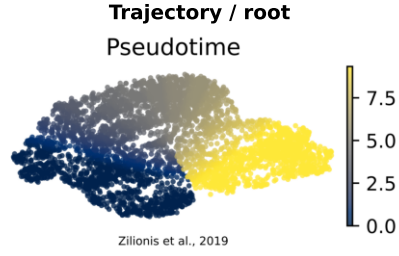
Single dominant Gast→pSom axis: gastrulation high at the origin declines along pseudotime (ρ=−0.58) while somitogenesis rises to a terminal readout (ρ=+0.77; pSom-high decile mean pt 0.99). Strong Gast↔pSom cross-repression (smoothed r=−0.53, near Puram −0.49) but joint-high enrichment 0.73 < 1 — mutually exclusive, one axis not a co-expression hub. FLAG: Crest is globally depleted and weakly U-shaped (high at BOTH tips, lowest mid; $t^2>0$, $p<0.001$) — not the reference terminal readout.

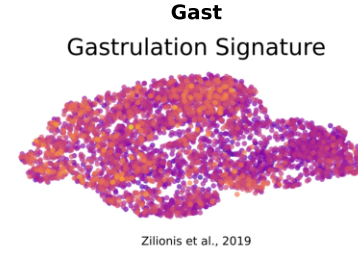
Gate 3 — hub mutual exclusivity → NA

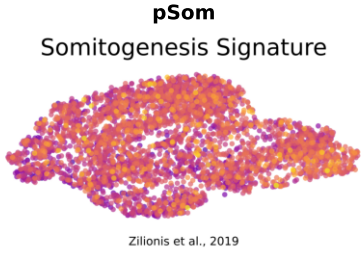
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity (the Wu 2021 organised-cross-repression non-branching case).

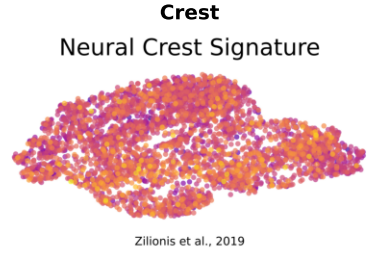
STEP FIGURES

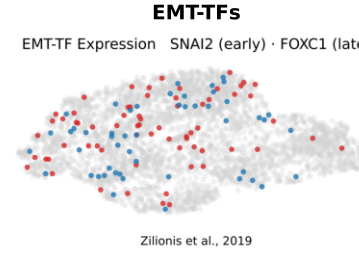












Dataset. NSCLC (Zilionis et al. 2019), inDrop — 7 patients / 31,179 cells (master scope); malignant only: 3,920 cells across 6 malignant-bearing samples (drop 1 sub-100 spillover sample, p2) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis's NSCLC malignant manifold is a single monotone gastrulation → somitogenesis axis with strong Gast↔pSom cross-repression (smoothed r=−0.53), but the decisive hub test shows the two are mutually exclusive (joint-high enrichment 0.73 < 1), so it is one axis, not Puram's co-expression hub — no bifurcation, and Gate 3 has no hub to test. The two density lobes are NOT a proliferative split (6.4% cycling; both lobes low) nor histological nor single-patient; the axis is cell-cycle-free (survives non-cycling cells, r=−0.53) so the whole manifold is kept. EMT-TFs are weak & root/early-loaded: SNAI2 (root) → FOXC1 (terminus) chosen by data over Puram's pair (FOXC1 recovered by widening the panel, the Kürten lesson), though end-separation is small (Δpt=0.02). Crest is the weak link: globally depleted and only shallowly U-shaped (high at both tips, not terminal) — flagged.

Parameter choices

Min cells/sample	≥131 (clean dominant cliff 232→75, r3.09; drops p2; 6/7)	**Geometry**	non-branching single axis (curve 2 tips, 0 forks, fold-free)
Integration	Harmony θ=10; PCA50; n_pcs=20; spread=1.3	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (ElPiGraph)
Island removal	radius-graph 6× median NN — 0 removed	**Scope**	whole manifold (no lobe scoping)
Root node	Gast-high / pSom-low curve tip 2	**EMT-TFs**	SNAI2 (early) → FOXC1 (late), 3% floor